THE INTEGRAL HODGE CONJECTURE FOR THE FERMAT SEXTIC FOURFOLD

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ABSTRACT. Let $X \subset \mathbb{P}^5$ be the Fermat sextic fourfold $x_0^6 + \cdots + x_5^6 = 0$. We show that every integral Hodge class of degree four on X is the class of an algebraic cycle, so the integral Hodge conjecture holds for X. The rational statement is due to Shioda, and Braun, Fortin, Lopez Garcia and Villaflor Loyola proved it after inverting 6. We compute the lattice S of integral primitive Hodge classes (rank 1751) with the vanishing-cycle method of Aljovin, Movasati and Villaflor Loyola. Linear subspaces together with the complete intersection surfaces of types $(1, 2, 3)$ and $(2, 2, 3)$ span a sublattice of index exactly 3^{540} in S. Three further explicit families lower the index to 3^{105}: surfaces in quadric-cone sections built from conics six times tangent to the Fermat sextic curve, pairs of such conics in all positions, and their twists by the Fermat group. The $105 = 15 \cdot 7$ remaining directions lie in the fifteen sectors of the sign group with four odd characters. We close them with algebraic correspondences built from Hodge endomorphisms of the Fermat sextic curve that are not combinations of automorphisms. Such endomorphisms exist because the Jacobian of the curve is isogenous to the tenth power of an elliptic curve with complex multiplication by $\mathbb{Z}[e^{2\pi i/3}]$. Through the inductive structure of Shioda and Katsura, these correspondences carry the fully algebraic sector with six odd characters onto every sector with four. The proof is computer-assisted and exact. The class of every cycle is computed from its equations over $\mathbb{Q}(e^{2\pi i/12}, 2^{1/3})$, and every lattice computation is either p-adic with an explicit precision bound or over $\mathbb{Z}$. Each computation was certified by an automated research system, with negative controls and recorded hashes, and a standalone verifier reproduces the decisive ones.

1. INTRODUCTION

For a smooth complex projective variety X, the *integral Hodge conjecture* in degree $2k$ asks whether the cycle class map

$$CH^k(X) \longrightarrow \mathrm{Hdg}^{2k}(X, \mathbb{Z}) = \{\alpha \in H^{2k}(X, \mathbb{Z}) : \alpha_{\mathbb{C}} \in H^{k,k}(X)\}$$

is surjective. It fails in general: for torsion classes [5], and for non-torsion classes on very general hypersurfaces of high degree in $\mathbb{P}^4$ [14]; see also [23, 10]. It holds in several cases of interest, for instance for cubic fourfolds [26] and for threefolds of Kodaira dimension zero [24].

For the Fermat varieties $X_m^n = \{x_0^m + \cdots + x_{n+1}^m = 0\}$, Shioda [21, 22] reduced the rational Hodge conjecture to a combinatorial problem on characters and proved it in many cases, among them X_6^4. Aljovin, Movasati and Villaflor Loyola [1] gave an algorithm that compares the lattice spanned by linear cycles with the lattice of integral Hodge classes, written in vanishing cycles; with it they confirmed the integral Hodge conjecture for the quartic and quintic Fermat fourfolds. Degtyarev and Shimada [11] showed that the linear subspaces of X_m^4 span a primitive sublattice of H_4 for $3 \leq m \leq 12$. For the sextic fourfold, linear cycles do not even span the rational Hodge classes. Braun, Fortin, Lopez Garcia and Villaflor Loyola [8] exhibited a rational basis of algebraic classes (1001 linear cycles, 720 surfaces of Aoki–Shioda type and 30 of a third type)

Date: 1 November 2026.

2020 *Mathematics Subject Classification.* Primary 14C30; Secondary 14C25, 14J70, 14H40, 14K22.

Key words and phrases. Integral Hodge conjecture, Fermat varieties, algebraic cycles, Pham cycles, Jacobians with complex multiplication, correspondences, computer-assisted proof.

and proved the Hodge conjecture with coefficients in $\mathbb{Z}[1/6]$. The integral statement remained open.

Theorem 1.1. *Let $X = X_6^4 \subset \mathbb{P}^5$ be the Fermat sextic fourfold. Every class in $\mathrm{Hdg}^4(X,\mathbb{Z})$ is the class of an algebraic cycle with integral coefficients. Consequently the integral Hodge conjecture holds for X in all degrees.*

The proof has two parts: a computation that measures exactly how far the known cycles are from $\mathrm{Hdg}^4(X,\mathbb{Z})$, and a new family of algebraic classes that closes the gap.

Theorem 1.2 (The index chain)**.** *Let $S \subset H^4(X,\mathbb{Z})$ be the lattice of integral primitive Hodge classes (rank 1751), and let $A_0 \subset S$ be the primitive part of the lattice spanned by the 3240 planes of X and by the complete intersection surfaces of Aoki–Shioda type $(1,2,3)$ and $(2,2,3)$ (Section 3).*

(a) *$[S : A_0] = 3^{540}$; in particular $A_0 \otimes \mathbb{Z}_{(\ell)} = S \otimes \mathbb{Z}_{(\ell)}$ for every prime $\ell \neq 3$.*
(b) *Adding the classes of surfaces in quadric-cone sections (Section 3.3) lowers the index to 3^{345}; adding classes of pairs of six-tangent conics in all positions (Section 3.4), to 3^{120}; adding classes of their twists by the Fermat group (Section 3.5), to 3^{105}.*
(c) *With A' the lattice reached in* (b)*, an explicit finite list of 5611 classes (Section 3.5), $S/A' \cong \bigoplus_s (\mathbb{Z}/3)^7$, the sum over the fifteen sectors s of the sign group with exactly four odd characters (Section 2.5).*

The last $105 = 15 \cdot 7$ directions are reached by a *transport* (Section 5). Let C be the Fermat sextic curve. Its Jacobian is isogenous to E^{10} with $E = \mathbb{C}/\mathbb{Z}[\omega]$ [13, 2, 6], so $\mathrm{End}(J(C))$ is much larger than the span of the automorphisms of C. On the Pham lattice $H_1(C^\circ,\mathbb{Z}) \cong \mathbb{Z}^{25}$ of the affine curve $C^\circ = \{x^6 + y^6 = 1\}$, the integral endomorphisms that preserve the Hodge type and commute with the twist $(x,y) \mapsto (\zeta x, \zeta y)$ form a lattice $\mathcal{E}$ of rank 85, of which the automorphisms span only 45 (Proposition 4.2). For any pair of coordinates $\{x_p, x_q\}$, the inductive structure of Shioda and Katsura presents X as birational to $(C \times Y)/\mu_6$. Through it, every $\beta \in \mathcal{E}$ gives an algebraic correspondence $\mathcal{T}_\beta$ on X that acts as $\pm\beta \otimes 1$ on the characters with $a_p + a_q \neq 6$ (Proposition 5.4). Taken through the even pair of a sector with four odd characters, these correspondences carry the sector with six odd characters, which is fully algebraic, onto that sector.

Theorem 1.3. *For every sector s with four odd characters and even pair $\{x_p, x_q\}$,*

$$S_s \otimes \mathbb{Z}_{(3)} \;=\; \Big(A'_s + \sum_{\beta \in \mathcal{E}} \mathcal{T}_\beta \big(S_{\mathrm{hub}}\big)_s\Big) \otimes \mathbb{Z}_{(3)}.$$

What is new. All earlier constructions of algebraic cycles on Fermat varieties are linear subspaces, joins of lower-dimensional cycles, or surfaces of Aoki–Shioda type and their relatives [21, 19, 3, 8]. Every one of them is built from the automorphism group of the Fermat variety and from complete intersections. The classes that close the gap come instead from correspondences on the Fermat curve that are *not* induced by automorphisms. Their existence is a consequence of complex multiplication, and their role was suggested by the symmetry structure of the residual classes rather than found by search. Section 8 explains how the idea arose.

Computer assistance. The lattice computations (the Hodge lattice S, the indices of Theorem 1.2, and the transport in Theorem 1.3) were carried out with the research system KAIROS. Each computation is submitted as a contract: inputs are read by hash, checks are evaluated on the program's own outputs, and the record keeps the hashes of the program and of the inputs, the outputs and the result of every check. None of the computations the proof uses involves

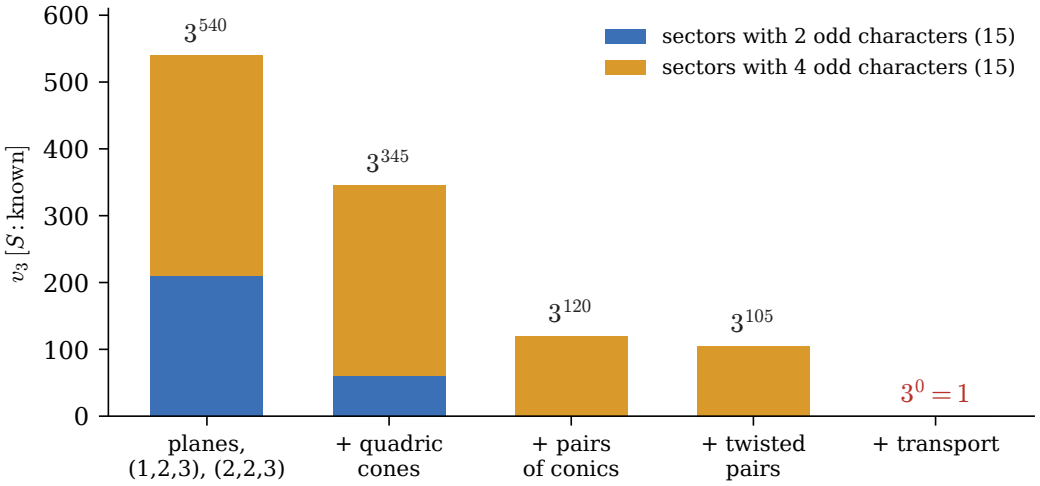


FIGURE 1. The 3-adic valuation of the index of the known algebraic classes in S, stage by stage, split into the fifteen sectors with two odd characters and the fifteen with four (the sectors with zero or six odd characters never miss anything). The transport of Section 5 closes the last 105 directions.

rounding. The classes of the cycles are computed from their equations in exact arithmetic (Lemma 2.3), and the lattices 3-adically with the margin of Lemma 2.1. The decisive step was also certified through periods, with rounding, and the two computations agree. The certificates and a standalone verifier are described in Appendix A. The theoretical part (Sections 4 and 5) uses only classical facts: the Lefschetz $(1,1)$ theorem, descent along a cyclic group, Pham's formula for periods, the theorem of Villaflor Loyola on the classes of complete intersections and the inductive structure of Fermat varieties.

2. Setup

2.1. **Characters.** Let $\zeta = e^{2\pi i/6}$ and $\omega = \zeta^2$. The group $G = \mu_6^6/\mu_6$ acts on X by scaling the coordinates, and S_6 permutes them. For $a = (a_0, \dots, a_5) \in \{1, \dots, 5\}^6$ with $\sum a_i \equiv 0 \pmod 6$, let $V(a) \subset H^4_{\text{prim}}(X, \mathbb{C})$ be the eigenspace on which $g \in G$ acts by $\prod g_i^{a_i}$. Then

$$H^4_{\text{prim}}(X, \mathbb{C}) = \bigoplus_a V(a), \qquad \dim V(a) = 1, \qquad V(a) \subset H^{4-q,q} \ \text{ with } q = \tfrac{1}{6}\textstyle\sum_i a_i - 1$$

[21]. There are 2605 characters, and those of Hodge type $(2,2)$ are the ones with $\sum_i a_i = 18$; there are 1751 of them. By Griffiths' residue theorem they correspond to the monomials x^{a-1} of degree 12 in the Jacobian ring of $\sum x_i^6$.

2.2. **Pham cycles.** The affine piece $U = X \cap \{x_0 \neq 0\}$ is the Milnor fibre $\{x_1^6 + \cdots + x_5^6 = 1\}$. By Pham [18], $H_4(U, \mathbb{Z})$ is free with basis the cycles δ_β, $\beta \in \{0, \dots, 4\}^5$, and for the form $\omega_a = x^{a-1}\, dx_1 \wedge \cdots \wedge dx_5/df$, $a \in \{1, \dots, 5\}^5$,

$$\int_{\delta_\beta} \omega_a = \frac{1}{6^5} \frac{\prod_{i=1}^5 \Gamma(a_i/6)}{\Gamma(\sum_{i=1}^5 a_i/6)} \prod_{i=1}^5 \left(\zeta^{a_i(\beta_i+1)} - \zeta^{a_i\beta_i}\right) \tag{1}$$

(see [17, Ch. 15]). Pham's basis is a tensor product: $H_4(U, \mathbb{Z}) = L^{\otimes 5}$ with $L = \mathbb{Z}[t]/(1+t+\cdots+t^5)$, the Pham lattice of the zero-dimensional Fermat variety $\{x^6 = 1\}$; this is the Sebastiani–Thom isomorphism [20], and [16] describes the resulting integral Hodge structure. The map $H_4(U, \mathbb{Z}) \to$

$H_4(X,\mathbb{Z}) \cong H^4(X,\mathbb{Z})$ is onto the primitive lattice $h^\perp$, and its kernel is spanned over $\mathbb{Q}$ by the characters with $\sum_{i\geq 1} a_i \equiv 0 \pmod 6$. The character $a = (a_1,\dots,a_5)$ of U corresponds to the character $(a_0,a_1,\dots,a_5)$ of X with $a_0 \equiv -\sum_{i\geq 1} a_i$.

2.3. **The integral Hodge lattice.** The intersection form of $H_4(U,\mathbb{Z})$ in the basis δ_β is explicit [1]. A class in $H_4(U,\mathbb{Z})$ maps to a Hodge class exactly when its periods (1) vanish on the characters of type $(4,0)$ and $(3,1)$. Following [1], we computed the image S of this kernel p-adically for $p = 2, 3$, the only primes that matter (Proposition 3.2). We used Smith elimination over $\mathbb{Z}/p^K$, tracking the unimodular row operations. The following elementary lemma says when such a finite computation yields the integral kernel.

Lemma 2.1 (Finite precision)**.** *Let M be an $n \times m$ matrix over $\mathbb{Z}_p$ of rank r over $\mathbb{Q}_p$. Suppose that Smith elimination modulo p^K, tracking the row operations U, finds r pivots, all of valuation $\leq v < K$. Then the $n-r$ rows of U that were not used as pivots, lifted to $\mathbb{Z}_p$, are congruent modulo p^{K-v} to a $\mathbb{Z}_p$-basis of the left kernel $\{y : yM = 0\}$, which is saturated.*

Proof. Lift U to $\mathbb{Z}_p$; it stays unimodular. Each non-pivot row u satisfies $uM = p^K w$, and uM lies in the row space of M, whose elementary divisors are at most p^v. Hence $p^K w = yM$ with $y \in p^{K-v}\mathbb{Z}_p^n$, and $u - y$ lies in the kernel. These corrected rows, together with the r pivot rows, still form a basis of $\mathbb{Z}_p^n$ modulo p, so the corrected rows span the saturated kernel. □

For the constraint matrix of S (the period numerators of the 427 forms of F^3 on the cycles δ_β, written in the basis $1,\zeta$ of $\mathbb{Z}[\zeta]$, a 3125×854 integer matrix), the elimination finds 854 pivots, the rank over $\mathbb{Q}$, with valuations at most 3 at both primes; we used $K = 15$ at 3 and $K = 24$ at 2. So the computed kernel, of rank $2271 = 3125 - 854$, is the saturated kernel modulo 3^{12} (and modulo 2^{21}), more than enough for everything below, which is used only modulo 3 or modulo 3^{12}. At 3 this was rechecked by an independent program from the definition of S: the 2271 stored rows satisfy the constraints modulo 3^{15} and are independent modulo 3, and the constraint matrix has Smith valuations $0^{11}1^{205}2^{526}3^{112}$.

Computation 2.2. The lattice S has rank 1751. The Smith invariants of its Gram matrix are $0^{900}1^{109}2^{640}3^{100}5^2$ at 2 and $0^{1043}1^{281}2^{281}3^{145}4^1$ at 3 ($\text{exponent}^{\text{multiplicity}}$). Two independent sets of conditions (the 427 forms of F^3 and all 854 forms of type $\neq (2,2)$) give the same lattice.

2.4. **Classes of complete intersections.** Let $Z = \{f_1 = f_2 = f_3 = 0\} \subset X$ be a complete intersection surface and choose forms g_j with $\sum_i x_i^6 = \sum_j f_j g_j$. For a Hodge character a let $\Phi_Z(a)$ be the coefficient of $\prod_i x_i^{5-a_i}$ in the Jacobian determinant $\det \partial(f_1,f_2,f_3,g_1,g_2,g_3)/\partial(x_0,\dots,x_5)$. By Villaflor Loyola [25], the periods of $[Z]$ against the residue forms ω_a of $x^{a-1}\Omega/F^3$ are $\Phi_Z(a)$ times a universal constant. For $c \in \mathbb{Q}^{3125}$ write $\hat{c}(a) = \sum_\beta c_\beta \zeta^{a\cdot\beta}$, $a \in \{1,\dots,5\}^5$; these are the coordinates of c on the characters.

Lemma 2.3 (Exact bridge)**.** *Let c be Pham coordinates of $[Z]$, defined modulo the radical. Then $\hat{c}(a) = 0$ for the characters of type other than $(2,2)$, and on the Hodge characters, up to a sign that is the same for all Z,*

$$\hat{c}(a) = \frac{\Phi_Z(a)}{\kappa\,\alpha(a)\prod_{i=1}^5(\zeta^{a_i}-1)}, \qquad \kappa = \tfrac{3}{2},$$

$$\alpha(a) = 2^{\frac{-n_1+3n_4+4n_5}{3}}\,3^{\frac{n_1-n_4-n_5}{2}}\,\sin\tfrac{\pi a_0}{6}\;e^{\pi i(a_1+\cdots+a_5)/6},$$

where n_j is the number of $i \in \{0,\dots,5\}$ with $a_i = j$.

Proof. The first statement holds because $[Z]$ is a Hodge class. In the chart $x_0 = 1$ we have $F = 1 + f$, $f = \sum_{i\geq 1} x_i^6$, and the residue of $x^{a-1}\Omega/F^3$ is, up to a universal factor, the second

derivative in t at $t=-1$ of the Gelfand–Leray form $x^{a-1}dx/df$ on $\{f=t\}$ [17, Ch. 15]. By homogeneity its integral over the transported cycle $\delta_\beta(t)$ is $t^{S/6-1}$ times (1), $S=a_1+\cdots+a_5$, so

$$\int_{\delta_\beta}\omega_a=\mu\,\frac{\prod_{i=1}^5\Gamma(a_i/6)}{\Gamma(S/6)}\Big(\frac{S}{6}-1\Big)\Big(\frac{S}{6}-2\Big)e^{\pi iS/6}\prod_{i=1}^5\big(\zeta^{a_i(\beta_i+1)}-\zeta^{a_i\beta_i}\big)$$

with μ universal. Since $S=18-a_0$, $\Gamma(S/6)=(2-\frac{a_0}{6})(1-\frac{a_0}{6})\Gamma(1-\frac{a_0}{6})$, and the reflection formula [12, 5.5.3] turns the Gamma factor into $\frac{1}{\pi}\sin\frac{\pi a_0}{6}\prod_{i=0}^5\Gamma(\frac{a_i}{6})$. Reflection and duplication [12, 5.5.3, 5.5.5] give $\Gamma(\frac{j}{6})=\pi^{(j-2)/2}\,c_j\,\Gamma(\frac13)^{e_j}$ with $(c_j,e_j)=(2^{-1/3}3^{1/2},2)$, $(1,1)$, $(1,0)$, $(2\cdot3^{-1/2},-1)$, $(2^{4/3}3^{-1/2},-2)$ for $j=1,\dots,5$. On a Hodge character $\sum_i(a_i-3)=0$, so the powers of $\Gamma(\frac13)$ cancel and $\prod_{i=0}^5\Gamma(\frac{a_i}{6})=\pi^3\,2^{(-n_1+3n_4+4n_5)/3}3^{(n_1-n_4-n_5)/2}$. So the periods are $\mu\pi^2\alpha(a)\prod_i(\zeta^{a_i}-1)\,\zeta^{a\cdot\beta}$, and comparing with Villaflor's theorem gives the formula with a universal constant κ. The class is real, so $\hat c(-a)=\overline{\hat c(a)}$; this fixes κ up to sign, and the self-intersection $[P_0]^2_{\mathrm{prim}}=21-\frac16$ of a plane gives $\kappa^2=\frac94$. □

The intersection matrix of the cycles δ_β is $\Psi=V^{\otimes5}+(V^{\mathsf T})^{\otimes5}$, with V the 5×5 matrix with 1 on the diagonal and -1 above it [18]. The lemma is confirmed by exact identities that do not involve any surface: with λ_a the eigenvalue of Ψ on the character a and $\rho(a)=1/(\kappa\alpha(a)\prod_i(\zeta^{a_i}-1))$, we have $-6^7\lambda_a\rho(a)\rho(-a)=1$ for all 1751 Hodge characters, which says that the intersection form transported by the lemma is the residue pairing $-6^{-7}\sum_a\Phi_Z(a)\Phi_{Z'}(-a)$ [9]. That pairing gives $Z^2-(\deg Z)^2/6$, namely $125/6$, 60 and 96, for all 12960 surfaces P_1, all 6480 surfaces P_2 and the planes we checked. The lemma turns the class of a surface into exact integers: $\hat c$ lies in $\frac12\mathbb{Z}[\zeta]$ (in $\mathbb{Z}[\zeta]$ for $[Z]_0$), although $\Phi_Z(a)$ lies in $\mathbb{Q}(\zeta_{12},2^{1/3})$, and the pairing vector $v=\Psi c=(\Psi(G^{-1})^{\otimes5})\hat c$, with $G=(\zeta^{ab})_{1\le a\le5,\,0\le b\le4}$, is the integer vector $(\langle[Z],\delta_\gamma\rangle)_\gamma$. The group acts on the functionals by the change of variables in the determinant: $\Phi_{t(Z)}(a)=\prod_i u_i^{a_i}\,\Phi_Z(a)$ for $t=\mathrm{diag}(u_i)$, $u_i^6=1$, and $\Phi_{\sigma(Z)}(\sigma a)=\mathrm{sgn}(\sigma)\Phi_Z(a)$ for a permutation σ of the coordinates.

2.5. **Sectors.** The sign group $\Sigma=\mu_2^6/\mu_2\cong(\mathbb{Z}/2)^5\subset G$ has order 32, prime to 3. Hence

$$S\otimes\mathbb{Z}_{(3)}=\bigoplus_\varepsilon S_\varepsilon,\qquad \pi_\varepsilon=\tfrac{1}{32}\textstyle\sum_{\sigma\in\Sigma}\varepsilon(\sigma)\sigma^*,$$

the sum over the characters ε of Σ, and $32\pi_\varepsilon$ maps S into S. A character a lies in the sector given by its parity vector $(a_i \bmod 2)$, which has an even number of odd entries: there are $1+15+15+1=32$ sectors, containing 20, 36, 70 and 141 Hodge characters for 0, 2, 4 and 6 odd entries. A sector with four odd entries is determined by its two even coordinates $\{x_p,x_q\}$, its *even pair*. The sector with six odd entries is called the *hub*. We write a sector as a binary word in $x_1,\dots,x_5$ (with 1 for odd); for example, 00111 has even pair $\{x_1,x_2\}$ and odd coordinates x_0,x_3,x_4,x_5.

Projections commute with algebraicity: π_ε is a rational combination of automorphisms, so $\mathrm{Alg}\otimes\mathbb{Z}_{(3)}=\bigoplus_\varepsilon\mathrm{Alg}_\varepsilon$ for the lattice $\mathrm{Alg}\subset S$ of primitive algebraic classes.

3. Known algebraic cycles and the index chain

3.1. **Planes and Aoki–Shioda surfaces.** For a permutation σ of the coordinates and $\eta_j^6=-1$, the planes

$$\{x_{\sigma(0)}=\eta_1x_{\sigma(1)},\ x_{\sigma(2)}=\eta_2x_{\sigma(3)},\ x_{\sigma(4)}=\eta_3x_{\sigma(5)}\}$$

give $15 \cdot 6^3 = 3240$ linear cycles. If $\lambda^3 = 2\varepsilon$ with $\varepsilon = \pm 1$, then on $\{x_a^2 = \lambda x_b x_c\}$ we have $x_a^6 + x_b^6 + x_c^6 = (x_b^3 + \varepsilon x_c^3)^2$. Hence X contains the complete intersections

$$P_1 = \{x_a = \eta x_b,\ x_c^2 = \lambda x_d x_e,\ x_d^3 + \varepsilon x_e^3 = \iota x_f^3\}, \qquad \eta^6 = -1,\ \iota = \pm i,$$
$$P_2 = \{x_a^2 = \lambda x_b x_c,\ x_d^2 = \lambda' x_e x_f,\ x_b^3 + \varepsilon x_c^3 = \iota(x_e^3 + \varepsilon' x_f^3)\},$$

of types $(1,2,3)$ and $(2,2,3)$ (12960 and 6480 surfaces); compare [3, 4, 8].

3.2. **Convention for classes of surfaces.** An effective surface Z has $[Z] \cdot h^2 = \deg Z > 0$, so $[Z]$ is not primitive. We fix a plane $P_0 \subset X$ and put

$$[Z]_0 = [Z] - (\deg Z)\,[P_0].$$

Since $P_0 \cdot h^2 = 1$, $[Z]_0$ is integral, primitive and Hodge, so $[Z]_0 \in S$. We do not use the orthogonal projection $[Z] - \frac{\deg Z}{6} h^2$, which has denominators. All lattices of algebraic classes below are spanned by classes $[Z]_0$. In a sector s with ε nontrivial, $32\pi_s[P_0] = \sum_\sigma \varepsilon(\sigma)(\sigma^*[P_0] - [P_0])$ is a combination of classes $[P]_0$ of planes, so $\pi_s[Z]_0 \equiv \pi_s[Z]$ modulo $A_0 \otimes \mathbb{Z}_{(3)}$.

Let A_0 be the lattice spanned by the classes $[Z]_0$ of the planes and of the surfaces P_1, P_2. Equivalently, A_0 consists of the combinations $\sum c_i[Z_i]$ with $\sum c_i \deg Z_i = 0$, and it lies in S.

Computation 3.1. The Smith invariants of the Gram matrix of A_0 are $0^{900}1^{109}2^{640}3^{100}5^2$ at 2 and $0^{338}1^{671}2^{566}3^{145}4^{31}$ at 3. The classes $[Z]_0$ of 1001 planes, 720 surfaces P_1 and 30 surfaces P_2 form a $\mathbb{Z}_3$-basis of A_0. The 3-adic Smith form of their Gram matrix was reproduced independently with the Singular code of [8].

Proposition 3.2. $[S : A_0] = 3^{540}$. *In particular* $A_0 \otimes \mathbb{Z}_{(\ell)} = S \otimes \mathbb{Z}_{(\ell)}$ *for every prime* $\ell \neq 3$.

Proof. Since $A_0 \subset S$ have the same rank, $\operatorname{disc}(A_0) = \operatorname{disc}(S)\,[S : A_0]^2$. Let A be the lattice spanned by h^2 and all the cycles. We chose a subset of 1752 generators of A that generates it modulo 5, and computed the determinant of its Gram matrix exactly over $\mathbb{Z}$: it is $2^{1698} \cdot 3^{2483}$. So $\operatorname{disc}(A)$ involves only the primes 2 and 3. Since $\mathbb{Z}h^2 \oplus A_0 \subset A$ has index dividing 6, the same holds for $\operatorname{disc}(A_0)$, and therefore for $[S : A_0]$. At 2, the valuations of $\operatorname{disc}(A_0)$ and $\operatorname{disc}(S)$ are both 1699 (Computations 2.2 and 3.1), so $2 \nmid [S : A_0]$. At 3 they are 2362 and 1282, so $v_3[S : A_0] = (2362 - 1282)/2 = 540$. □

3.3. **Surfaces in quadric-cone sections.** Let $W = X \cap \{x_a^2 = \lambda x_b x_c\}$ with $\lambda^3 = 2\varepsilon$. On W we have $x_a^6 + x_b^6 + x_c^6 = u^2$ with $u = x_b^3 + \varepsilon x_c^3$. So the rational map $\pi\colon W \dashrightarrow \mathbb{P}(3,1,1,1)$, $x \mapsto (u : x_d : x_e : x_f)$, takes values in the K3 surface

$$K = \{u^2 + x_d^6 + x_e^6 + x_f^6 = 0\} \subset \mathbb{P}(3,1,1,1),$$

the double plane branched along the Fermat sextic curve. The map π is a morphism outside the six points of W with $x_d = x_e = x_f = 0$. Its fibre over a general point $(u_0 : d_0 : e_0 : f_0)$ of K is the part $\{z \neq 0\}$ of the curve

$$\{x_a^2 = \lambda x_b x_c,\ x_b^3 + \varepsilon x_c^3 = u_0 z^3\} \subset \mathbb{P}^3_{(x_a : x_b : x_c : z)}, \qquad (x_d, x_e, x_f) = z\,(d_0, e_0, f_0).$$

This curve is a smooth complete intersection of type $(2,3)$, hence of genus 4 by adjunction; the six points with $z = 0$ are removed. Pulling back divisors along the morphism π (rational functions pull back to rational functions) and taking closures in W gives a homomorphism

$$\Phi\colon \operatorname{Pic}(K) = \operatorname{NS}(K) \longrightarrow \operatorname{Cl}(W) \longrightarrow H^4(X, \mathbb{Z}),$$

whose image consists of classes of algebraic surfaces. The known curves of K (the 36 rational curves over the tritangent lines, the 36 curves of type $(2,3)$ and the cubics) span a sublattice of index 3^4 in $\operatorname{NS}(K)$. The new classes are realized by explicit surfaces

$$\{x_a^2 = \lambda x_b x_c,\ q(x_d, x_e, x_f) = 0,\ x_b^3 + \varepsilon x_c^3 = g(x_d, x_e, x_f)\},$$

where q is a conic six times tangent to the Fermat sextic curve and g is a cubic with $x_d^6 + x_e^6 + x_f^6 + g^2 \in (q)$. For example $q = x_d^2 + x_e^2 + x_f^2$ and $g = \pm i\sqrt{3}\, x_d x_e x_f$ work, by the identity $p^3 + r^3 + s^3 - 3prs = (p + r + s)(\dots)$ applied to $p = x_d^2$, $r = x_e^2$, $s = x_f^2$.

Computation 3.3. For each of the 360 quadric cones, let $\Gamma_1, \dots, \Gamma_4$ be generators of $\mathrm{NS}(K)$ modulo the known curves. The 1440 primitive classes $\Phi(\Gamma_i)_0$ (Section 3.2) lie in S, and the lattice they span together with A_0 has index 3^{345} in S. For the cone $x_0^2 = \lambda x_1 x_2$, the 60 explicit surfaces above also give four new directions modulo A_0.

3.4. **Pairs of conics.** If C_1, C_2 are conics six times tangent to the Fermat sextic curve, with cubics g_1, g_2 as above, then $x_0^6 + x_1^6 + x_2^6 \equiv -g_1^2$ on $C_1(x_0, x_1, x_2) = 0$ and $x_3^6 + x_4^6 + x_5^6 \equiv -g_2^2$ on $C_2(x_3, x_4, x_5) = 0$. So X contains

$$\{C_1(x_0, x_1, x_2) = 0,\ C_2(x_3, x_4, x_5) = 0,\ g_1 = i\, g_2\}.$$

From the 30 symmetric conics that are not of cone type we get 900 surfaces, and 720 permutations move them to all positions.

Computation 3.4. With the pairs in all positions the index drops to 3^{120}; the sectors with two odd characters close completely.

3.5. **Twisted pairs.** The symmetric conics are not stable under the twists $x_k \mapsto \omega x_k$, so the images of the pairs under $(\mu_3)^6 \rtimes S_6$ are new surfaces.

Computation 3.5. Let $\mathcal{G}$ be a fixed list of 2880 elements of $(\mu_3)^6 \rtimes S_6$: four twists, drawn once with a recorded seed, for each of the 720 permutations. Adding the images of the pairs under the elements of $\mathcal{G}$ lowers the index to 3^{105}, one direction in each sector with four odd characters.

We define A' as the lattice spanned by the following explicit list of 5611 integral classes. Each is an integer combination of classes $[Z]_0$ of the surfaces above, with coefficients drawn once from recorded seeds; the verifier of Appendix A reproduces the list.

- 1771 combinations, with coefficients in $\{0, 1, 2\}$, of the classes of the 22680 planes and surfaces P_1, P_2;
- the 1440 classes $\Phi(\Gamma_i)_0$ of the quadric cones;
- 600 combinations of the images of the 900 pairs of conics under the 720 permutations, and 1800 combinations of their images under the elements of $\mathcal{G}$.

All of them are classes of algebraic cycles, so $A' \subset \mathrm{Alg}$, and each is computed exactly from the equations of its surfaces by Lemma 2.3. The first 1771 already span $A_0 \otimes \mathbb{Z}_{(3)}$, since their index in S is $3^{540} = [S : A_0]$. The proof uses only two facts about A': that $A' \subset \mathrm{Alg}$, and the index $[S : A'] = 3^{105}$. It never uses exhaustiveness of the search; the full orbits of the pairs may well span a larger lattice.

3.6. **The residual spaces.** For a sector s with four odd characters, let $W_s = S_s/(A'_s + 3S_s) \cong \mathbb{F}_3^7$. Three of its seven directions are spanned by the Hodge classes

$$\tfrac{1}{3} \textstyle\sum_{j=0}^{2} \tau^j([P_1] - \sigma[P_1]),$$

where $\sigma\colon x_c \mapsto \omega x_c$ changes the cube root in P_1 and τ permutes (x_d, x_e, x_f) cyclically. These *weight-six classes* are integral but were not known to be algebraic; call their span R_s. The other four directions form the permutation module of the four triangles $\{x_p, x_q, x_k\}$ through the even pair, twisted by the sign character. The following fact is the one the transport needs.

Computation 3.6. In the sector 00111 (even pair $\{x_1, x_2\}$), every class of S_s whose periods vanish outside the characters with $a_1 + a_2 = 6$ maps to 0 in W_s. This concerns the full saturated sublattice $(S_s \cap V_{\mathrm{cone}}) \otimes \mathbb{Z}_{(3)}$: it was computed as the kernel, over $\mathbb{Z}/3^{12}$, of the exact period numerators (in $\mathbb{Z}[\zeta]$) on the other characters, for a $\mathbb{Z}_3$-basis of S_s.

stage	each of the 15 sectors with 2 odd characters	each of the 15 sectors with 4 odd characters	$v_3[S:\text{known}]$
planes, $(1,2,3)$, $(2,2,3)$	14	22	540
+ quadric-cone surfaces	4	19	345
+ pairs of conics	0	8	120
+ twisted pairs	0	7	105
+ transport (Section 5)	0	0	0

TABLE 1. Missing directions per sector, stage by stage. The sector with no odd character and the hub (six odd characters) miss nothing already for A_0.

4. Hodge endomorphisms of the Fermat sextic curve

4.1. **The Pham lattice of the curve.** Let $C^\circ = \{x^6 + y^6 = 1\}$ and let $C \subset \mathbb{P}^2$ be its closure, a smooth curve of genus 10 with six points at infinity. As above, $H_1(C^\circ, \mathbb{Z}) = L \otimes L \cong \mathbb{Z}^{25}$, and the twists $(x, y) \mapsto (\zeta^i x, \zeta^j y)$ split $H_1(C^\circ, \mathbb{C})$ into lines $V(a, b)$, $a, b \in \{1, \dots, 5\}$. The character (a, b) is *holomorphic* if $a + b < 6$ (the form $x^{a-1}y^{b-1}dx/y^5$ is holomorphic on C), *cusp* if $a + b = 6$, and *antiholomorphic* if $a + b > 6$. Let $u = a + b \bmod 6$; the twist $g\colon (x, y) \mapsto (\zeta x, \zeta y)$ acts on $V(a, b)$ by ζ^u. The loops around the points at infinity span the saturated sublattice $K_\infty = H_1(C^\circ, \mathbb{Z}) \cap \bigoplus_{a+b=6} V(a, b)$, and $H_1(C, \mathbb{Z}) = H_1(C^\circ, \mathbb{Z})/K_\infty$.

Definition 4.1. Let $\mathcal{E} \subset \operatorname{End}_{\mathbb{Z}} H_1(C^\circ, \mathbb{Z})$ be the lattice of endomorphisms that map each *block* $V_\tau = \bigoplus_{(a,b)\in\tau} V(a, b)$ into itself, where τ runs over the nine classes of characters with the same type and the same u.

Proposition 4.2. *$\mathcal{E}$ has rank $85 = 30+30+25$ (holomorphic, antiholomorphic and cusp blocks). The twists and the swap $(x, y) \mapsto (y, x)$ span a sublattice of rank 45, saturated at 3. The remaining forty directions link characters (a, b) and (a', b') of the same block with $(a', b') \notin \{(a, b), (b, a)\}$; see Fig. 2.*

Proof. $\mathcal{E} \otimes \mathbb{C} = \bigoplus_\tau \operatorname{End}(V_\tau)$, of dimension $\sum_\tau |\tau|^2$, and the blocks have sizes $1, 2, 3, 4$ (holomorphic), $4, 3, 2, 1$ (antiholomorphic) and 5 (cusp). The twists span the diagonal endomorphisms, and the swap composed with twists spans the entries $(a, b) \to (b, a)$; together they give $2 \cdot 25 - 5 = 45$ dimensions. We computed an integral basis of $\mathcal{E}$ exactly, as the kernel of the linear conditions over $\mathbb{Q}(\zeta)$, reduced it by LLL (all entries lie in $\{-3, \dots, 3\}$), and computed the index of the automorphism span in its saturation. □

Remark 4.3. These endomorphisms exist because $J(C)$ is isogenous to E^{10} with E the elliptic curve with complex multiplication by $\mathbb{Z}[\omega]$ [13, 2, 6]: $\operatorname{End}(J(C)) \otimes \mathbb{Q} \cong M_{10}(\mathbb{Q}(\omega))$, while the automorphisms of C span a much smaller subalgebra. For degrees prime to 6, integral endomorphisms of Fermat Jacobians outside the group ring were studied in [15].

Lemma 4.4. *Let $\beta \in \mathcal{E}$. Then β preserves K_∞ and induces an endomorphism $\bar\beta$ of $H_1(C, \mathbb{Z})$ that preserves the Hodge decomposition. There is a divisor D_β on $C \times C$ whose correspondence action on $H_1(C, \mathbb{Z})$ is $\bar\beta$ and which is invariant, as a cycle, under $g \times g$.*

Proof. β preserves the cusp block, hence the saturated lattice K_∞, and $\bar\beta$ maps holomorphic characters to holomorphic ones, so $\bar\beta \in \operatorname{End}(J(C))$. Since $\operatorname{End}(J(C)) = \operatorname{Pic}(C \times C)/(p_1^* \operatorname{Pic} C + p_2^* \operatorname{Pic} C)$ [7, Thm. 11.5.1], $\bar\beta$ is induced by a divisor D; this is the Lefschetz $(1, 1)$ theorem on $C \times C$. Since $\bar\beta$ commutes with g, the correspondence $(g \times g)D - D$ acts trivially on H_1. So its class is a combination of $[\mathrm{pt} \times C]$ and $[C \times \mathrm{pt}]$, and it vanishes, since $g \times g$ preserves both classes

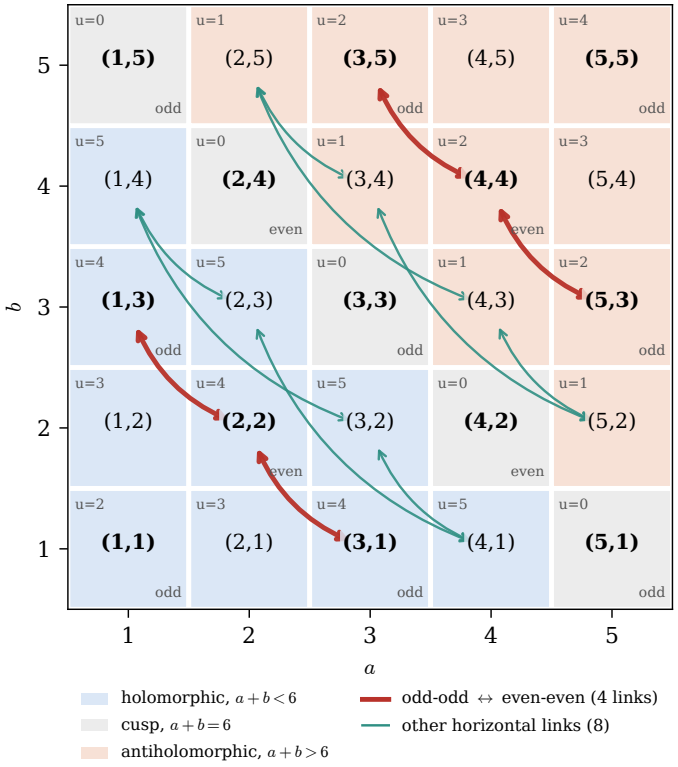


FIGURE 2. The 25 characters (a, b) of the affine Fermat sextic curve, coloured by type, with $u = a + b \bmod 6$. Arcs join characters linked by an endomorphism in $\mathcal{E}$ that is not a combination of automorphisms (eight more links among the cusp characters are not drawn). The four red links join an odd-odd character to an even-even one off the cusps; they carry the hub into a sector with four odd characters.

and intersection numbers, and these two classes have a non-degenerate intersection matrix. Thus the class c of D in $\mathrm{NS}(C \times C)$ is invariant.

Let $L = \mathcal{O}(D)$. Then $(g\times g)^*L\otimes L^{-1} \in \mathrm{Pic}^0(C\times C) = \mathrm{Pic}^0(C)^2$. The map $Q \mapsto (g\times g)^*Q\otimes Q^{-1}$ is $(g^* - 1) \times (g^* - 1)$ there, and it is surjective because g has no fixed vectors on $H^1(C)$ (every character of C has $u \neq 0$). So, after twisting by some $Q \in \mathrm{Pic}^0$, L is $(g \times g)$-invariant. It is linearizable because $H^2(\mathbb{Z}/6, \mathbb{C}^*) = 0$. For an ample linearized A and $n \gg 0$, take eigen-sections s of $L \otimes A^n$ and s' of A^n; then $D_\beta = \mathrm{div}(s) - \mathrm{div}(s')$ is an invariant divisor with class c. □

5. The transport

5.1. **The join.** Fix a pair $\{p, q\}$ and let R be the other four coordinates. Put

$$C = \{x_p^6 + x_q^6 + t^6 = 0\} \subset \mathbb{P}^2, \qquad Y = \Big\{ \sum_{r\in R} x_r^6 = t'^6 \Big\} \subset \mathbb{P}^4,$$

and

$$\varphi\colon C \times Y \dashrightarrow X, \qquad ([x_p : x_q : t], [x_R : t']) \longmapsto [t'x_p : t'x_q : t\, x_R].$$

It is invariant under the diagonal action $(t, t') \mapsto (\zeta t, \zeta t')$ of μ_6 and generically the quotient by it: this is the inductive structure of Shioda and Katsura [22, Thm. I]. Its indeterminacy locus is $B = \{t = 0\} \times \{t' = 0\}$, the six points at infinity of C times $Y_\infty = Y \cap \{t' = 0\}$. Let

$\widetilde{W} = \mathrm{Bl}_B(C \times Y)$. The diagonal μ_6 acts on both normal directions of B with the same weight, so it fixes the exceptional divisor E pointwise. Hence $\widetilde{W}/\mu_6$ is smooth, and it is isomorphic to the blow-up of X along the six points $\{x_p^6 + x_q^6 = 0,\ x_R = 0\}$ and the surface $X \cap \{x_p = x_q = 0\}$ [22, Thm. I]. We write $\tilde{\varphi}\colon \widetilde{W} \to X$ for the resulting morphism. On C, the auxiliary twist $t \mapsto \zeta t$ acts as g^{-1}.

Definition 5.1. For $\beta \in \mathcal{E}$, with D_β as in Lemma 4.4, let $\widetilde{Z}_\beta \subset \widetilde{W} \times \widetilde{W}$ be the strict transform of $D_\beta \times \Delta_Y$, and

$$\mathcal{T}_\beta = \tfrac{1}{6}\,(\tilde{\varphi} \times \tilde{\varphi})_* \widetilde{Z}_\beta.$$

Lemma 5.2. *$\mathcal{T}_\beta$ is an algebraic cycle with integral coefficients on $X \times X$. On rational cohomology, $\tilde{\varphi}^* \circ \mathcal{T}_{\beta *} = \widetilde{Z}_{\beta *} \circ \tilde{\varphi}^*$ on the horizontal characters defined below.*

Proof. The group $\mu_6 \times \mu_6$ acts generically freely on every component of $\widetilde{Z}_\beta$ that $\tilde{\varphi} \times \tilde{\varphi}$ does not contract, because μ_6 acts freely on $Y \setminus Y_\infty$. The cycle $\widetilde{Z}_\beta$ is invariant, with multiplicities, under the diagonal $\Delta\mu_6$, because D_β is $(g \times g)$-invariant as a cycle (Lemma 4.4). Let a component have stabilizer $\mathrm{St} \subset \mu_6 \times \mu_6$. Its $\Delta\mu_6$-orbit has $k = 6/|\mathrm{St} \cap \Delta|$ components, all with the same image. That image is therefore counted $k\,|\mathrm{St}| = 6\,[\mathrm{St} : \mathrm{St} \cap \Delta]$ times. Contracted components push forward to 0. Hence $(\tilde{\varphi} \times \tilde{\varphi})_* \widetilde{Z}_\beta$ is divisible by 6 as a cycle; integrality of $\tilde{\varphi}^*$ is never used. Over $\mathbb{Q}$, $W_* = \pi_* Z_* \pi^*$ for $W = (\pi \times \pi)_* Z$, and $\pi^*\pi_* = \sum_{h \in \mu_6} h^*$ on the quotient part. The correction terms, which come from the difference between strict and total transforms and from the exceptional classes of the blow-ups, carry only characters with $a_p + a_q \equiv 0$ or $a_p = a_q = 0$. So they do not meet the horizontal characters. □

5.2. **The horizontal action.** Call a character χ of X *horizontal* for $\{p, q\}$ if $a_p + a_q \neq 6$, and a *cone character* if $a_p + a_q = 6$. Horizontal characters are those of $[H^1(C) \otimes H^3(Y)]^{\mu_6}$; cone characters come from the exceptional loci.

Lemma 5.3 (Affine join)**.** *Assume $p, q \geq 1$, let $R' = R \setminus \{0\}$ and work in the chart $x_0 = 1$. Let*

$$C_{\mathrm{af}} = \{u_p^6 + u_q^6 = 1\} \subset \mathbb{C}^2, \qquad G = \Big\{ \sum_{r \in R'} w_r^6 + \tau^6 = 1 \Big\} \subset \mathbb{C}^4,$$

an affine piece of C and of Y (τ the auxiliary coordinate), and $\psi\colon C_{\mathrm{af}} \times G \to U$, $\psi(u, (w, \tau)) = (\tau u, w)$; up to sixth roots of -1 in the coordinates, ψ is the restriction of φ, and it avoids E. Let γ be a 1-cycle on C_{af}, w a Pham 2-cycle of $\{\sum_{R'} w_r^6 = 1\}$ and $\theta = \sum_k c_k [\zeta^k]$ a 0-chain on $\{\tau^6 = 1\}$. Then

$$\psi_*(\gamma \times (w * \theta)) = \pm\,(\theta(g)\gamma) * w,$$

where $*$ is the join of Sebastiani–Thom, $\theta(g) = \sum_k c_k g^k$, and the sign does not depend on γ, w, θ.

Proof. The join $w * \theta \subset G$ is swept by $((1-s)^{1/6} w,\ s^{1/6}\zeta^k)$, $s \in [0,1]$, and $\psi(u, ((1-s)^{1/6} w, s^{1/6}\zeta^k)) = (s^{1/6}\zeta^k u,\ (1-s)^{1/6} w)$, which sweeps the join of $g^k\gamma$ and w in U. □

Proposition 5.4. *Assume $p, q \geq 1$ and use Pham coordinates in the chart $x_0 = 1$. For every $y \in H^4(X, \mathbb{Q})$ and every horizontal character χ, the χ-components of $\mathcal{T}_\beta(y)$ and of $\pm(\beta \otimes 1)(y)$ coincide, with a sign independent of y and χ. Here β acts on the two Pham axes of x_p, x_q and the identity on the others.*

Proof. By Sebastiani–Thom, the Pham basis of U is the join of the Pham basis of C_{af} with that of $\{\sum_{R'} w_r^6 = 1\}$, so it suffices to treat $y = \iota_*(v * w)$ with v in the eigenspace of g of index $u \neq 0$ ($\iota\colon U \to X$). Take $\theta = e_u = \sum_k \zeta^{-ku}[\zeta^k]$; then $\theta(g)$ is 6 on that eigenspace and 0 on the others, so Lemma 5.3 gives $\iota_*(v * w) = \pm\frac{1}{6}\psi_*(v \times (w * e_u))$. By Lemma 5.2, $\mathcal{T}_{\beta *}\psi_* = \psi_* \widetilde{Z}_{\beta *}$ on these classes,

and $\widetilde{Z}_{\beta*}(v\times y') = (\bar\beta v)\times y'$. The class βv is a lift of $\bar\beta v$ in the same eigenspace, since β preserves u and the lift is unique up to cusp classes, which have $u = 0$. Hence $\mathcal{T}_{\beta*}(v * w) = \pm(\beta v) * w$ up to terms with $u = 0$. These terms vanish on horizontal characters because $\mathcal{T}_\beta$ commutes with the diagonal twist of (x_p, x_q) and therefore preserves u. □

Proposition 5.5. *For every $x \in S$, the class $\delta = \mathcal{T}_\beta(x) - (\beta\otimes 1)(x)$ has no horizontal components, and $\delta_s \in S_s \otimes \mathbb{Z}_{(3)}$ for every sector s.*

Proof. Replacing β by $-\beta \in \mathcal{E}$ if necessary, the sign in Proposition 5.4 is $+$. Both operators commute with the twists of the coordinates in R, so they preserve the R-part of characters and hence $a_p + a_q \bmod 6$; in particular they preserve the splitting into horizontal and cone characters. On horizontal characters they agree by Proposition 5.4, up to the exceptional classes of Lemma 5.2. These are supported on the cones $x_q = \eta x_p$, where the diagonal twist of (x_p, x_q) is isotopic to the identity, so they only have cone characters. Finally, $\mathcal{T}_\beta(x)$ is an integral Hodge class, $(\beta \otimes 1)(x)$ is integral in the Pham lattice and Hodge (it preserves the characters with $\sum a_i = 18$), and projections to nontrivial sectors kill h^2 and have denominator 32. □

Theorem 5.6. *Let s be a sector with four odd characters and even pair $\{p, q\} \subset \{1, \dots, 5\}$. If*

$$A'_s + \langle (\beta \otimes 1)(x)_s : \ \beta \in \mathcal{E},\ x \in S_{\mathrm{hub}} \rangle = S_s \otimes \mathbb{Z}_{(3)}, \tag{2}$$

then $\mathrm{Alg}_s = S_s \otimes \mathbb{Z}_{(3)}$.

Proof. Let $W'_s = S_s/(\mathrm{Alg}_s + 3S_s)$, a quotient of W_s. A permutation of the coordinates takes the sector 00111 to s, the pair $\{1, 2\}$ to $\{p, q\}$, S to S and algebraic classes to algebraic classes. So Computation 3.6 shows that classes of S_s supported on cone characters of $\{p, q\}$ vanish in W'_s. The hub is fully algebraic (Table 1), so for $x \in S_{\mathrm{hub}}$ the class $\mathcal{T}_\beta(x)$ is algebraic and $\mathcal{T}_\beta(x)_s \in \mathrm{Alg}_s$. By Proposition 5.5, $(\beta \otimes 1)(x)_s = \mathcal{T}_\beta(x)_s - \delta_s$ with δ_s of cone type, so $(\beta \otimes 1)(x)_s$ vanishes in W'_s. By (2) these classes and $A'_s \subset \mathrm{Alg}_s$ span W'_s, so $W'_s = 0$, that is $\mathrm{Alg}_s + 3S_s = S_s$. Since S_s/Alg_s is a finite 3-group, Nakayama's lemma gives $\mathrm{Alg}_s = S_s \otimes \mathbb{Z}_{(3)}$. □

6. The computation

Condition (2) of Theorem 5.6 was checked by two independent programs: through periods, with rounding, and exactly modulo 3. The exact check is the one the proof uses. It was run on the exact lattice A' of Section 3.5, whose generators come from their equations through Lemma 2.3. In all 32 sectors this lattice coincides with the one the period computation used, which had been obtained through floating-point periods.

6.1. **Through periods.** The sources are the classes $x = 32\,\pi_{\mathrm{hub}}(g)$, where g runs over the 2271 generators of $S \otimes \mathbb{Z}_{(3)}$ produced in Section 2.3; at 3 they generate S_{hub}. To keep memory bounded we replace them by 320 random combinations with coefficients in $\{-1, 0, 1\}$, and check that these generate the same lattice of the hub (rank 141, the same 3-adic valuation). For each of the 85 basis elements β of $\mathcal{E}$, we compute the periods of $(\beta\otimes 1)(x)$ in the sector s by contracting the Pham tensor on the two axes of the even pair. We then pair the result with a basis of S_s and round to integers. The index of A'_s plus the transported classes is computed by incremental Hermite elimination over $\mathbb{Z}/3^K$ and Smith forms. The controls are the following.

- *Known cases.* The twist of x_1 and the swap of (x_1, x_2), taken through the same code, act on the periods of X as the automorphisms do (deviations $1.4 \cdot 10^{-15}$ and $3.1 \cdot 10^{-16}$), and both lie in $\mathcal{E}$.
- *Negative control.* A random integral 25×25 matrix outside $\mathcal{E}$ gives non-integral pairings (deviation 0.5).
- *Integrality.* All pairings of the transported classes with the bases of the ten sectors are within 0.046 of an integer.

6.2. **Exactly, modulo 3.** By Nakayama's lemma, (2) only concerns the images in $S_s/3S_s$, so it can be checked without periods and without rounding. On the Pham lattice the twist of x_i acts on the i-th factor L by the matrix T of multiplication by t, with $Te_j = e_{j+1}$ for $j < 4$ and $Te_4 = -\sum_j e_j$. Hence the projection to the sector with parity vector s (of $a_1, \dots, a_5$) is

$$32\,\pi_s = \prod_{i=1}^{5} (1 + (-1)^{s_i} T_i^3),$$

the transport is $\beta \otimes 1$ on the two axes of the even pair, and the pairing coordinates of a class c are $\pm(32\,\pi_s c)^{\mathsf{T}}\Psi$, where Ψ is the integral 3125×3125 intersection matrix of the Pham cycles. Everything is integral, and we compute modulo 3. The pairing map $S_s \to \mathbb{Z}^{3125}$ is injective, and the generators of S come from Lemma 2.1. Since S_s has rank 70, the check that their pairing coordinates have rank 70 modulo 3 shows that the induced map $S_s/3S_s \to \mathbb{F}_3^{3125}$ is injective. Hence ranks computed modulo 3 in pairing coordinates are ranks in $S_s/3S_s$, which is exactly what Nakayama's lemma needs. The checks are as follows.

- *Conventions.* In every sector, the exact projections of 300 random combinations of the generators of S span exactly the stored S_s (rank 70 modulo 3), and those of 400 planes and complete intersections fall in A'_s.
- *Null control.* The twist and the swap of the even pair carry nothing from the hub into s.
- *Negative control.* A random integral matrix outside $\mathcal{E}$ carries the hub out of S_s.
- *Sources.* The 160 sources span the hub (rank 141), as the 2271 generators and the stored lattice do.
- *Before and membership.* A'_s has rank 63 modulo 3 in each sector (seven missing directions), and all transported classes lie in S_s.

6.3. **Results.** Both computations give the same answer. In each of the ten sectors whose even pair lies in $\{x_1, \dots, x_5\}$, A'_s has rank 63 and A'_s plus the transported classes has rank $70 = \operatorname{rank} S_s$ modulo 3, so condition (2) holds and the missing directions go from 7 to 0. In the sector 00111, the transport through the even pair $\{x_1, x_2\}$ reaches all seven directions of W_s, four of them outside R_s. Through the odd pairs $\{x_3, x_4\}$, $\{x_3, x_5\}$, $\{x_4, x_5\}$ it reaches six, but there Theorem 5.6 does not apply, because cone characters of an odd pair do carry directions (Fig. 3).

Remark 6.1. The exact certificate rounds only in one control, which the proof does not need: the 400 planes and complete intersections of the convention check are read from the floating-point bridge, whose Pham coordinates are rational because they are defined modulo the radical of Ψ; their pairings are integral to within $3 \cdot 10^{-13}$. In the exact computation of A' these surfaces are generators, so the control holds by construction.

7. Proof of Theorem 1.1

By Theorem 5.6 and the computation of Section 6, $\mathrm{Alg}_s = S_s \otimes \mathbb{Z}_{(3)}$ for the ten sectors with four odd characters and even pair in $\{x_1, \dots, x_5\}$. For the other five, a transposition $\sigma = (0\,k)$, with x_k odd in s, maps s to one of those ten. Since σ preserves S and algebraic classes, $\mathrm{Alg}_s = \sigma^{-1}\,\mathrm{Alg}_{\sigma s} = \sigma^{-1}(S_{\sigma s} \otimes \mathbb{Z}_{(3)}) = S_s \otimes \mathbb{Z}_{(3)}$. The sectors with zero, two and six odd characters are already reached by A' (Table 1). Hence $\mathrm{Alg} \otimes \mathbb{Z}_{(3)} = S \otimes \mathbb{Z}_{(3)}$. By Proposition 3.2, $A_0 \otimes \mathbb{Z}_{(\ell)} = S \otimes \mathbb{Z}_{(\ell)}$ for every prime $\ell \neq 3$. Thus every primitive integral Hodge class is algebraic.

For a plane $P \subset X$ we have $P \cdot h^2 = 1$. If $y \in \mathrm{Hdg}^4(X, \mathbb{Z})$ and $d = y \cdot h^2$, then $y - dP$ is primitive, so $\mathrm{Hdg}^4(X, \mathbb{Z}) = \mathbb{Z}[P] + S$ is algebraic. $H^*(X, \mathbb{Z})$ is torsion-free, $\mathrm{Hdg}^2(X, \mathbb{Z}) = \mathbb{Z}h$, and $\mathrm{Hdg}^6(X, \mathbb{Z}) \cong \mathbb{Z}$ is generated by the class of a line (Lefschetz). □

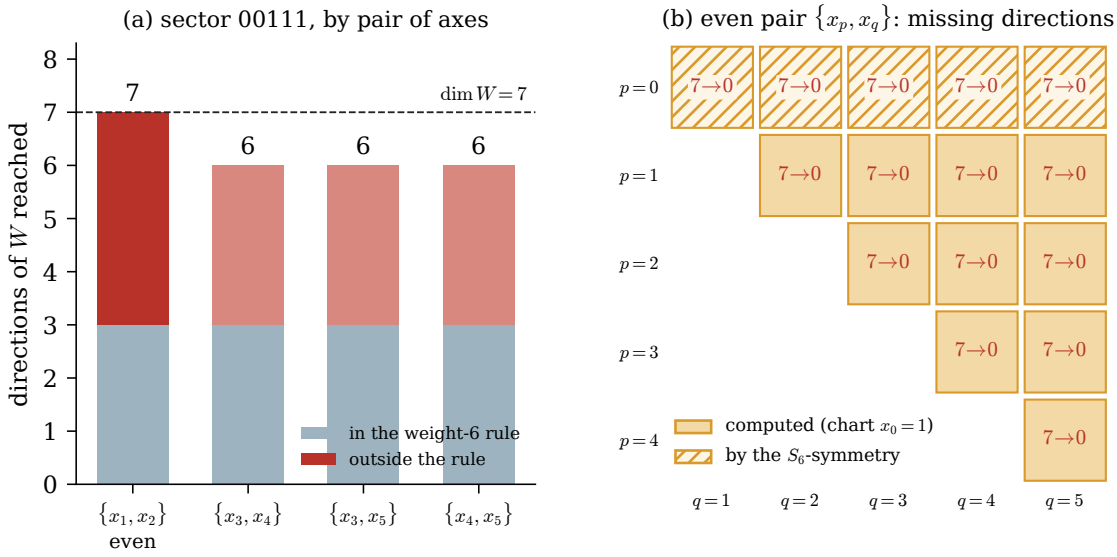


FIGURE 3. (a) Directions of W_s reached in the sector 00111 by transporting the hub through each pair of Pham axes; only the even pair counts for Theorem 5.6. (b) The fifteen sectors with four odd characters, indexed by their even pair $\{x_p, x_q\}$: ten were computed in the chart $x_0 = 1$, and the five with x_0 in the even pair follow by symmetry.

8. REMARKS AND QUESTIONS

(1) *Explicit cycles.* The divisors D_β are given by the Lefschetz $(1,1)$ theorem. They can be made explicit through elliptic quotients $q_1, q_2 \colon C \to E$ and endomorphisms ψ of E, as the correspondences $q_2^* \circ \psi \circ q_{1*}$. Writing equations for the surfaces $\mathcal{T}_\beta(x)$ would give an explicit set of generators of $\mathrm{Hdg}^4(X, \mathbb{Z})$.

(2) *Comparison with* [8]. [8, Thm. 2] bounds the denominators of integral Hodge classes in a basis $[A_i]_{\mathrm{prim}}$ by integers $d_i \in \{1, 2, 3, 6\}$ listed in their Table 1. With the order of indices inferred from their published procedures, the class of the surface $\{x_0^2 = \lambda x_1 x_2,\ x_3^2 + x_4^2 + x_5^2 = 0,\ x_1^3 + x_2^3 = \pm i\sqrt{3}\, x_3 x_4 x_5\}$ of Section 3.3 has denominator 3 at five indices where the table lists $d_i \in \{1, 2\}$; we computed this with their own code. This does not affect their $\mathbb{Z}[1/6]$ statement.

(3) *Other Fermat varieties.* The mechanism uses only that the Fermat curve of degree m has Hodge endomorphisms outside the span of its automorphisms that link characters of different parity. It should apply to the fourfolds X_m^4 with $6 \mid m$, where linear and Aoki–Shioda cycles may not suffice integrally.

(4) *How the idea arose.* After the families of Section 3 were exhausted, the residual directions were described in terms of their symmetry types. They are invariants of the diagonal action of μ_3 on products of Néron–Severi lattices that are not norms. Such classes can only be realized by an irreducible correspondence that is not a product. Following an analogy with isogenies of elliptic curves with complex multiplication, we tested the endomorphisms of the Fermat curve that are not automorphisms. Independent analogy searches in crystallography (coincidence lattices), topology (Smith theory and Massey products), chemistry (three-centre bonds) and lattice theory (Eisenstein glue codes) all pointed to the same kind of object.

Appendix A. Certificates, exactness and data

What a certificate is. Each computation was submitted to KAIROS as a contract. The contract fixes the program, the input files (identified by SHA-256), the declared outputs, and a list of checks: known cases computed by the same program (such as the twist and the swap, the self-intersection of a plane, or the reproduction of a stored lattice), negative controls, and invariants between outputs. The program runs in SageMath and the checks are evaluated on its outputs. The result is accepted only if every check passes; the record keeps the SHA-256 of the program and of the inputs, the outputs and the result of each check. A certificate is therefore a record of a checked run, not a proof by itself. Table 2 lists the main ones.

Exactness. The computations the proof uses are exact.

- $S \otimes \mathbb{Z}_{(3)}$: the constraint matrix is an integer matrix, and the kernel computed modulo 3^{15} is the saturated kernel modulo 3^{12} by Lemma 2.1. An independent program checks the stored basis from the definition of S. The basis has rank 2271 modulo 3 and satisfies the constraints modulo 3^{15}, and the Smith valuations of the constraint matrix are at most 3.
- A': each of its 5611 generators is computed from the equations of its surfaces, through the Villaflor functional over $\mathbb{Q}(\zeta_{12}, 2^{1/3})$ and Lemma 2.3. Its pairing vector Ψc comes out as an exact integer vector, with entries up to $3.1 \cdot 10^9$. The index chain of Theorem 1.2 and the inclusion $A' \subset S$ are then computed modulo 3^{12}. The bridge itself is checked by exact identities: the form of Ψ, the identity $-6^7 \lambda_a \rho(a)\rho(-a) = 1$ on all Hodge characters, the self-intersections 125/6, 60 and 96, and the action of the group against direct determinants.
- The transport step: integer arithmetic modulo 3 (Section 6).
- $\mathrm{disc}(A)$: an exact integer determinant.

Products of float64 matrices occur only on integers whose partial sums stay below 2^{53}, and each such bound is asserted in the code.

Earlier computations with rounding. Before the exact computation, the generators of A' had been obtained from floating-point periods and rounded; in each batch the largest distance to an integer was between 10^{-13} and $7 \cdot 10^{-4}$. The exact computation gives the same lattices A'_s in all 32 sectors. On a sample of 120 surfaces, the floating-point pairing vectors agree with the exact ones to $6.5 \cdot 10^{-14}$. They carry the opposite orientation on the planes, a sign convention of that computation that changes no lattice; in particular, all the degrees $\deg \Phi(\Gamma_i)$ are divisible by 3. The transport through periods also used rounding, and the proof does not depend on it.

Verifier and data. A standalone verifier runs, in plain SageMath, the three certified programs of the exact chain: the 3-adic basis of S, the exact A' with the index chain, and the transport. Its inputs are the 3-adic basis of S (2271×3125 residues modulo 3^{15}) and the data of $\mathrm{NS}(K)$ used for the cone classes. The matrix Ψ is generated from Pham's formula, and every input is checked against its SHA-256. The verifier and its inputs will be deposited with this paper.

Acknowledgements. The computations were designed and certified with KAIROS, an automated research system developed by the author. Parts of the code and of the text were prepared with the assistance of an AI model (Claude, Anthropic); all statements were checked against the certified computations.

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statement	job	checks	time
the exact chain			
$S \otimes \mathbb{Z}_{(3)}$ from its definition	j20261101-030814-5c533640	6	213 s
exact A': $3^{540} \to 3^{345} \to 3^{120} \to 3^{105}$	j20261101-030311-882581ff	17	56 min
transport, exactly mod 3 on the exact A': $7 \to 0$	j20261101-040126-0f2bcafd	9	210 s
earlier certificates			
$[S : A_0] = 3^{540}$; with cone classes 3^{345}	j20261030-210914-95bfe167	10	70 min
sector map at 3	j20261030-222946-2fb722f1	6	31 min
six-tangent conic surfaces in $S \setminus A_0$	j20261030-214326-6128a8b5	6	373 s
their reach in one cone	j20261030-215127-a1b53d51	7	1004 s
pairs of conics in all positions: 3^{120}	j20261031-005914-2fda5cc1	4	67 min
twisted pairs: 3^{105}	j20261031-052956-48bfe238	10	107 min
A_0 against the code of [8] (at 3)	j20261031-024140-5e88cad1	–	111 s
A_0 against the code of [8] (at 2)	j20261031-041154-8b2f45e2	7	272 s
cone characters (Computation 3.6)	j20261031-215549-6e1a08c9	5	179 s
transport through periods, ten sectors $7 \to 0$	j20261101-002858-fb0c8c55	6	571 s
transport exactly mod 3, ten sectors $7 \to 0$	j20261101-013224-153e7a26	9	213 s

SHA-256 of the certified programs of the exact chain:
$S \otimes \mathbb{Z}_{(3)}$ `c8a1baf54fdfaa40477c43c64e7b7dee1a0b47b4d5d8e49505f5468af0c6a07b`
exact A' `89ccb1843737fd7436ddea4a4bddb760e047328854b88816e558b5ddb6501010`
transport `42c9dd5521483715ea1673b9d051b49a1bff3a40eed307f553f7a4be4c74b310`
and of the two earlier transport certificates: through periods `4b106221da2990de27e4b89d796fbcd3 1732a14209471bf27d62ff9b438b39e9`, exact `5315534f153521fbad0f47053d8baf57 17c4b4f9dbc3635ea79afbe411e337f5`.

TABLE 2. Main certificates of the KAIROS memory service. The exact chain is what the proof uses; the earlier certificates, several of them through floating-point periods, gave the same indices.

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KAIROS RESEARCH PROJECT
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